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Centrality Analysis of Cryptocurrency Transaction Networks Based on the Elliptic Dataset

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4. Summary

Background

Industry Pain Points

Blockchain's decentralization facilitates illicit transactions and money laundering, making AML detection in crypto networks essential for financial supervision. Centralized key nodes can cause system-wide risk propagation via abnormal behaviors.

Research Gap

Tree-based models and GNN are mainstream AML approaches with different advantages in feature learning and graph adaptability, but their comparative evaluation in real blockchain transactions is still limited.

Technical Introduction: data processing

Our research leverages the Elliptic++ dataset (200k+ Bitcoin transactions, 230k+ edges) with a three-step processing pipeline:

Acquisition & Preprocessing

Obtain raw data, perform cleaning, deduplication, and handle missing values for quality assurance.

Network Construction

Model transactions as nodes and fund flows as directed edges to build a complete transaction graph.

Feature Extraction

Utilize 166 native features and compute network centrality metrics to enrich feature space.



Obtain
Data



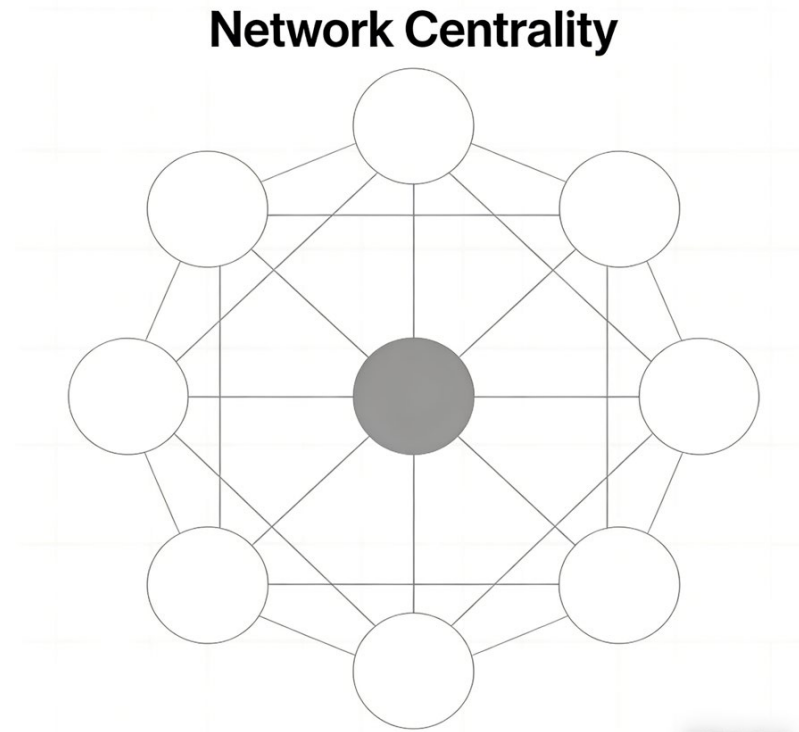
Cleaning and
Preprocessing



Build
Network

Technical Introduction: Centrality Metrics

- Degree centrality
- In-degree/Out-degree Ratio
- PageRank centrality
- Betweenness centrality
- Closeness Centrality
- Clustering Coefficient



Degree centrality

Nieminen (1974): Measure of centrality based upon degree.
For a network , degree centrality of node equals

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$$C_D(v) \frac{\deg(v)}{n - 1}$$

where n is the total number of nodes in the network

In-degree/Out-degree Ratio

Core Logic: Distinguish inward/outward capital flow in directed graphs.

Project Significance:

- High out-degree & low in-degree → Money laundering distribution end (early stage)
- High in-degree & low out-degree → Aggregation end / Darknet receiving address

PageRank centrality

PageRank centrality of node v_k :

$$C_{PR}(v_k) = \alpha \sum_{i=1}^n \frac{a_{k,i}}{d(v_i)} C_{PR}(v_i) + \beta$$

where α and β : constants

and $d(v_i)$: out-degree of node v_i if such degree is positive, or $d(v_i) = 1$ if the out-degree of v_i is null.

PageRank centrality

It follows that $\mathbf{C}_{PR}$ can be computed as:

$$\mathbf{C}_{PR} = (\mathbf{I} - \alpha \mathbf{D}^{-1} \mathbf{A})^{-1} \boldsymbol{\beta}$$

where α : damping factor

and $\boldsymbol{\beta}$: personalization vector

Both play the same role as in Katz centrality

Betweenness centrality

Core Logic: Nodes acting as "bridges" more frequently are more important. Measures the control over information flow.

Mathematical Expression: Ratio of shortest paths passing through the node in the entire network.

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

where

σ_{st} : Total number of shortest paths from node s to node t

$\sigma_{st}(v)$: Number of such paths passing through node v

Closeness Centrality

Core Logic: Measures the average distance from a node to all other nodes in the network.

Mathematical Expression: closeness centrality equals

$$C_C(v_k) = \frac{1}{\sum_{i=1}^n d(v_i, v_k)}$$

where $d(v_i, v_k)$: of edges in the geodesic linking v_i, v_k

Relative centrality based on closeness measure:

$$C'_C(v_k) = \frac{n - 1}{\sum_{i=1}^n d(v_i, v_k)}$$

Clustering Coefficient

Core Logic: Measures if a node's neighbors are connected to each other (clustering degree).

Project Significance:

- Illicit nodes (especially money laundering ones) have extremely low clustering coefficients.
- To avoid tracking, funds are transferred linearly instead of circulating among associates.
- An excellent indicator for identifying the "distribution-type" money laundering mode.

Technical Introduction: Model Training

From "Forest" to
"Gradient Boosting" & "Graph Structure"

- LightGBM (Leaf-wise)
- GraphSAGE GNN (F1-optimized parameters)
- GNN + Optuna hyperparameter search



LightGBM (Leaf-wise)

Background: Efficient GBDT implementation, core tree growth strategy, baseline model for blockchain AML detection

◆ Core Growth Logic

- Criterion: Max loss function reduction
- Split only the leaf node with the largest split gain
- Limit depth/leaf num to avoid overfitting

◆ Three Lightweight Core Technologies

- Histogram Optimization: Discretize continuous features, reduce memory/computation
- GOSS: Retain high-gradient samples, precise down-sampling
- EFB: Bundle mutually exclusive features, reduce high-dimensional feature dimensions

GNN — Graph Neural Network

- **Core idea:**

learn embeddings by aggregating features from a node and its neighbors.

- **GNN is a type of neural network for graph-structured data**, which consists of nodes and edges, such as blockchain transaction networks, fund flows, and social relations. Effectively captures structural patterns, relationships, and propagation paths, ideal for money laundering detection, fraud identification, and ring discovery.

- **Common models:**

GraphSAGE, GCN, GAT, GIN, etc.

GNN — Graph Neural Network

- **F1-optimized parameters**

The best set of model parameters optimized to maximize F1-score, for imbalanced tasks like AML detection.

- $$F_1 = 2 * \frac{Precision * Recall}{Precision + Recall}$$

- **Optuna hyperparameter search**

An automatic parameter searching process using Optuna to find the best hyperparameters.

Technical Introduction: Voting & Analysis

From "Simple Majority" to "Weighted Interpretation"

◆ Soft Voting

- Outputs probabilities instead of classes.
- Averages weighted probabilities from multiple models.
- More robust and accurate than hard voting.

◆ Shapley Additive Explanations

- Game theory-based model explanation method.
- Quantifies feature importance for each prediction.
- Improves transparency and interpretability.

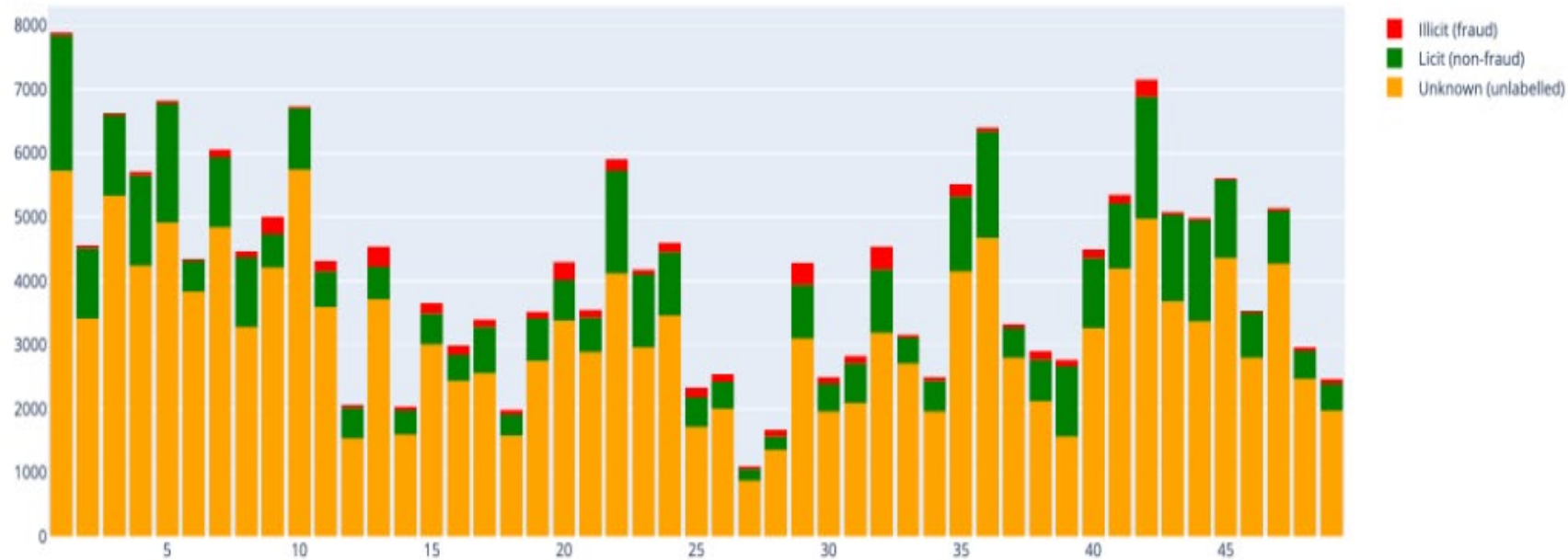
Results Analysis

- Data Structure
- Network Feature Analysis
- Model Architecture
- Performance Comparison

Results Analysis: Data Structure

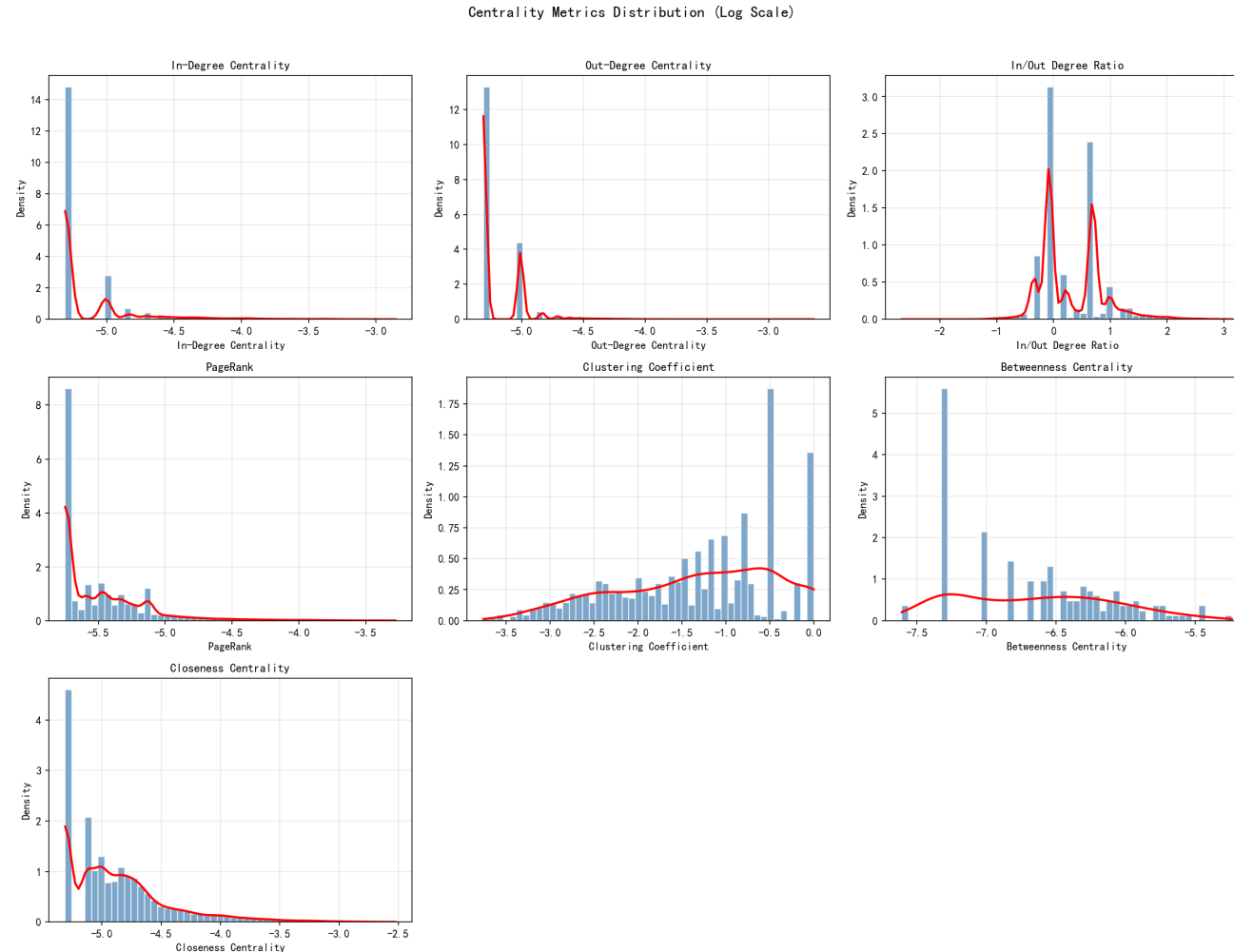
- **Data Structure**

The Elliptic dataset contains a total of 200k+ Bitcoin transactions, 230k+ edges, with three types of node labels: illicit, licit, and unknown. The distribution is shown in the figure



Results Analysis: Network Feature Analysis

- Centrality Histograms

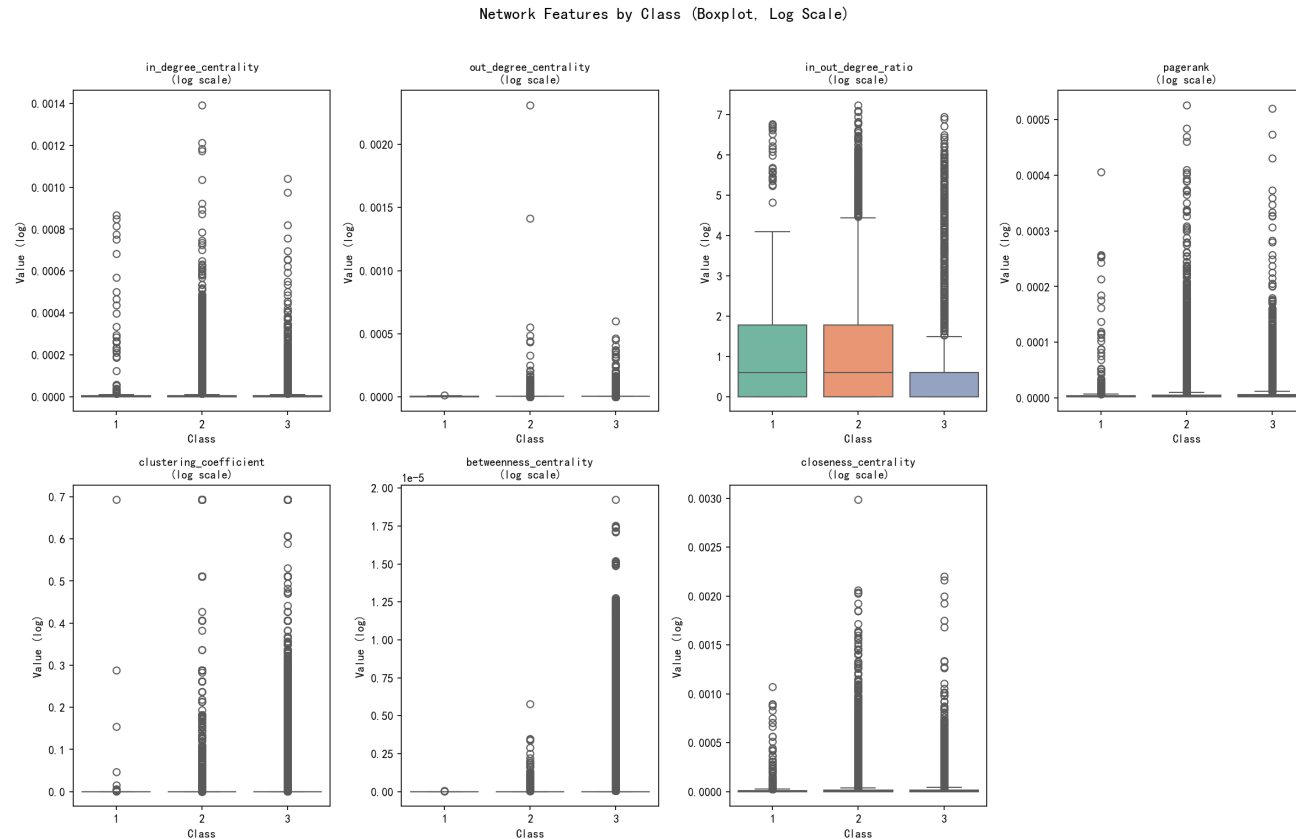


- Chart Interpretation

Exhibiting typical right-skewed long-tail distributions.

Results Analysis: Network Feature Analysis

- Boxplot of 7 network features grouped by class (log scale).



- Chart Interpretation**

Blue box = Unknown class (3), Green = Licit (2), Red = Illicit (1). Licit nodes show more dispersed feature value distributions, illicit nodes generally have lower and more concentrated feature values.

Results Analysis: Network Feature Analysis

- Centrality Statistics

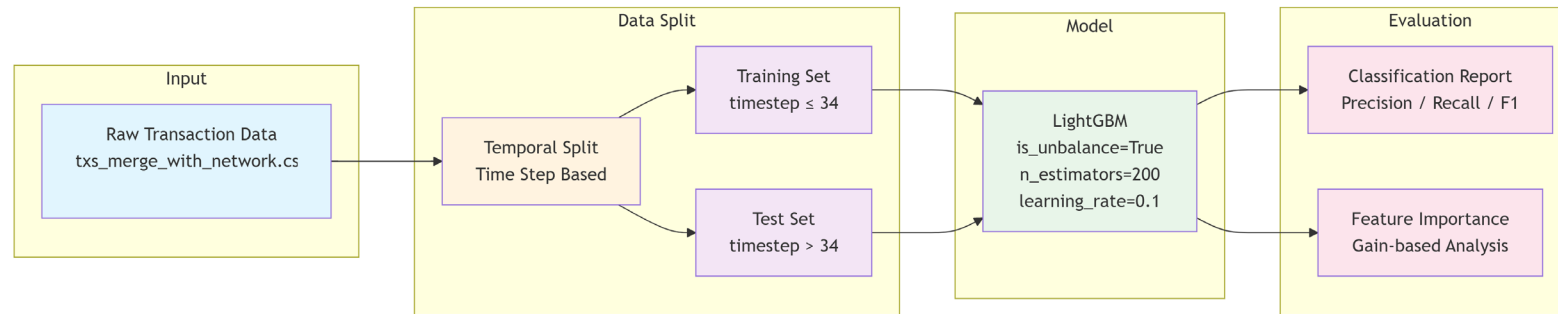
Metric	Licit (Mean/Median)	Illicit (Mean/Median)	Ratio
In-Degree Centrality	9.37e-06 / 4.91e-06	6.23e-06 / 4.91e-06	1.5
Out-Degree Centrality	5.82e-06 / 4.91e-06	3.64e-06 / 4.91e-06	1.6
In/Out Degree Ratio	4.83 / 0.83	4.33 / 0.83	1.12
PageRank	5.96e-06 / 2.57e-06	4.18e-06 / 2.61e-06	1.43
Clustering Coefficient	8.32e-03 / 0.00	3.56e-04 / 0.00	23.35
Betweenness Centrality	2.07e-09 / 0.00	2.16e-11 / 0.00	95.6
Closeness Centrality	2.11e-05 / 7.36e-06	1.13e-05 / 4.91e-06	1.88

◆ Key Findings

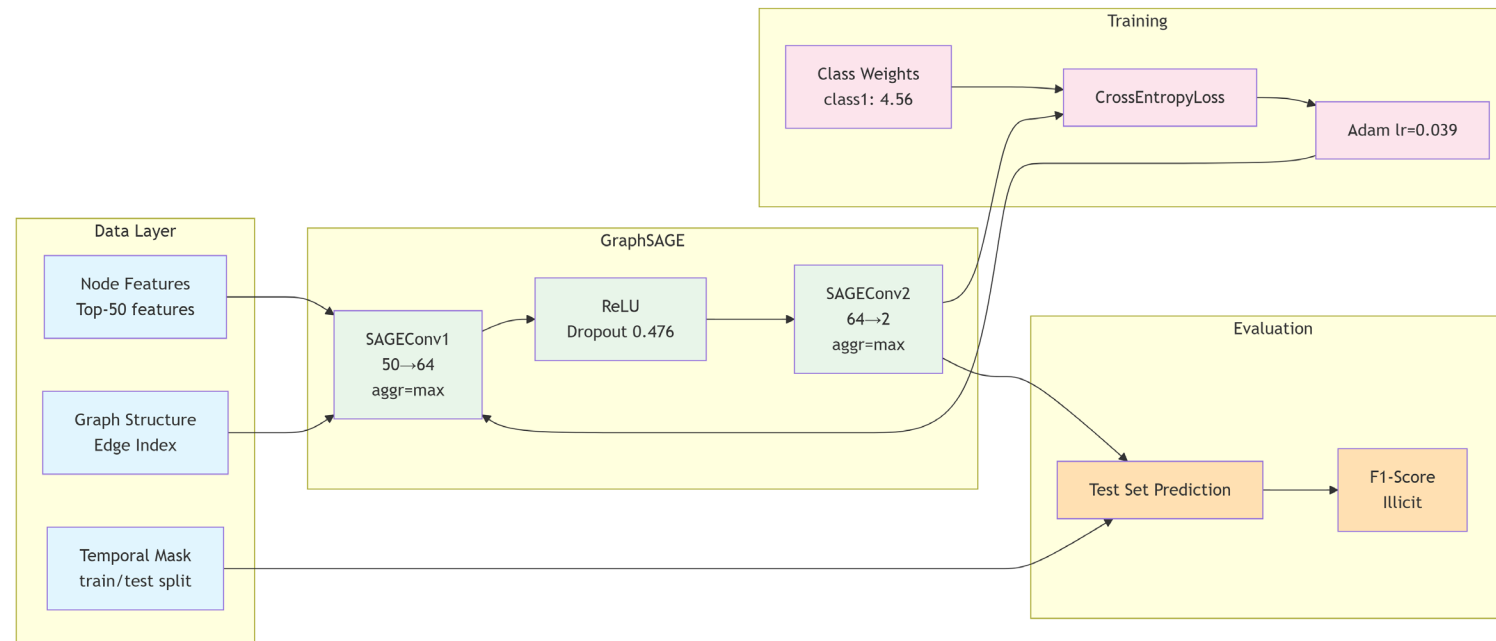
- Clustering Coefficient Difference:** Licit nodes have a clustering coefficient 23x higher than illicit nodes, indicating that licit transactions form tighter local communities (e.g., business networks).
- Betweenness Centrality Difference:** Licit nodes have betweenness centrality 95x higher than illicit nodes, suggesting licit nodes more often serve as network intermediaries (e.g., exchanges, payment platforms).

Results Analysis: Model Architecture

- LightGBM



- GraphSAGE



Results Analysis: Performance Comparison

◆ Key Finding

Tree-based models significantly outperform GNN on this task.

Model	Recall	Precision	F1-Score
LightGBM(189 features)	0.73	0.9	0.81
GraphSAGE(50 features, F1-optimized)	0.62	0.65	0.63
GraphSAGE(50 features, Recall-optimized)	0.87	0.16	0.27

◆ Key issues identified:

- **Unknown node interference:** 77% of nodes have unknown labels, contaminating message passing.
- **Temporal distribution shift:** Graph structure evolves over time.
- **Training strategy:** No validation set, no early stopping.

Summary

◆ Conclusion

In the current dataset, LightGBM performs significantly better than GraphSAGE, and simple and effective algorithms have advantages in specific scenarios.

◆ Future Prospects

- **Optimize training strategy:** Introduce validation set and early stop mechanism to prevent overfitting; Exploring Semi Supervised Learning to solve unknown node problems.
- **Advanced Model Exploration:** Attempting Time Graph Neural Networks to capture dynamic evolution.
- **Data and Feature Engineering:** Utilizing GAN to generate samples to alleviate class imbalance and explore deeper network topology feature information.
- **Hybrid model fusion:** Combining LightGBM feature importance with GNN structural information to construct a high-performance hybrid model with complementary advantages.